

PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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1. Hasil Running Code

Query :

```
CREATE TABLE hremployees (
  employee_id NUMBER(6) NOT NULL,
  first_name VARCHAR2(20),
  last_name VARCHAR2(25) NOT NULL,
  email VARCHAR2(25) NOT NULL,
  phone_number VARCHAR2(20),
  hire_date DATE NOT NULL,
  job_id VARCHAR2(10) NOT NULL,
  salary NUMBER(8,2),
  commission_pct NUMBER(2,2),
  manager_id NUMBER(6),
  department_id NUMBER(4),
  CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

-- INSERT data untuk department_id = 90 (Executive - dipakai soal no.4)
INSERT INTO hremployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003','DD-MON-YYYY'),
'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hremployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005','DD-MON-YYYY'),
17000, NULL, 100, 90);
INSERT INTO hremployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001','DD-MON-YYYY'),
17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY')
'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hremployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY')
'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hremployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY')
'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hremployees VALUES (124, 'Kevin', 'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY')
'ST_MAN', 5800, NULL, 100, 50);
INSERT INTO hremployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY')
'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hremployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-
'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hremployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-
'SA_REP', 11000, .3, 149, 80);
```

Output :

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTME
1	King	king@mail.com	5000	0,1	01-01-2023	
2	Kochhar	kochhar@mail.com	4500	(null)	15-02-2023	
3	Abel	abel@mail.com	3800	0,2	10-03-2023	
4	Taylor	taylor@mail.com	3200	0,15	05-04-2023	
5	Grant	grant@mail.com	2800	0,1	20-05-2023	
6	Hunold	hunold@mail.com	4000	(null)	20-01-2023	
7	Ernst	ernst@mail.com	3500	(null)	10-02-2023	
8	Lorentz	lorentz@mail.com	3000	(null)	01-03-2023	
9	Whalen	whalen@mail.com	2500	(null)	12-04-2023	
10	Zlotkey	zlotkey@mail.com	3100	0,1	01-05-2023	

2. Hasil Running Code 2

Query :

```
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in IN NUMBER
) IS
BEGIN
    FOR employee_rec IN (
        SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in
    ) LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + (emp.salary * increase_salary.increase_pct_in)
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
/

BEGIN
    increase_salary (90, .50);
    ROLLBACK;
END;
/

select * from employees where department_id = 90
```

Output :

	EMPLOYEE_ID	LAST_NAME	EMAIL		SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1	King	king@mail.com		5000	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com		4500	(null)	15-02-2023	

3. Jika Query Rollback dihapus

Maka yang terjadi Output dibagian salary akan bertambah karena sesuai dengan query increase_salary

Query :

```
BEGIN
  increase_salary (90, .50);
END;
/
```

Output :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPART
1	1	King	king@mail.com	7500	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com	6750	(null)	15-02-2023	